

Chapter 6 Part 2: Sexual Reproduction & Meiosis

- What are characteristics of the average cell?
- Meiosis = a unique type of cell division
- How is sex determined in offspring?
- What happens when meiosis goes wrong?

Corresponds with OpenStax Biology 2e Chapter 11

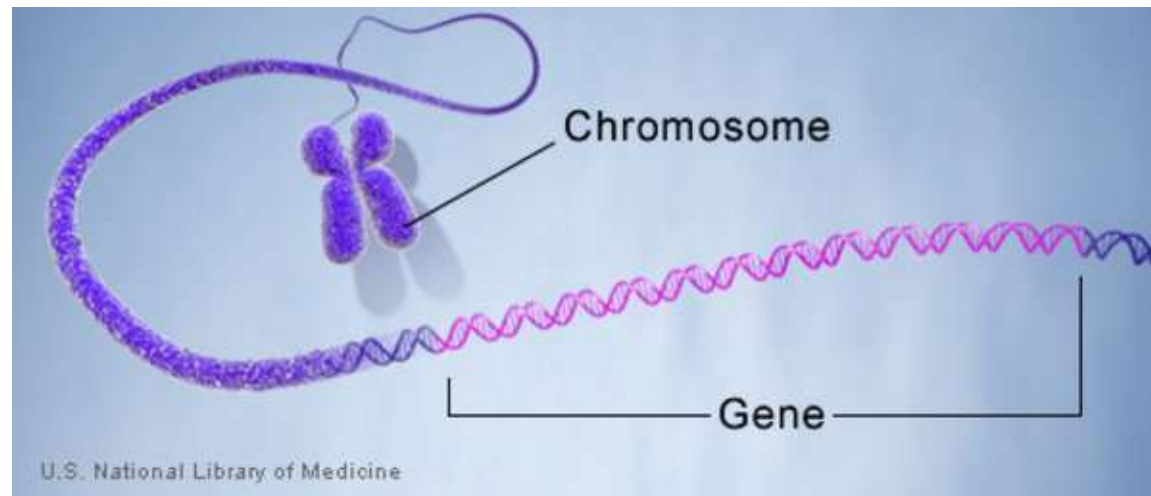
Learning Objectives

You will be able to:

1. Describe the behavior of chromosomes during meiosis
2. Describe the cellular events that take place during meiosis

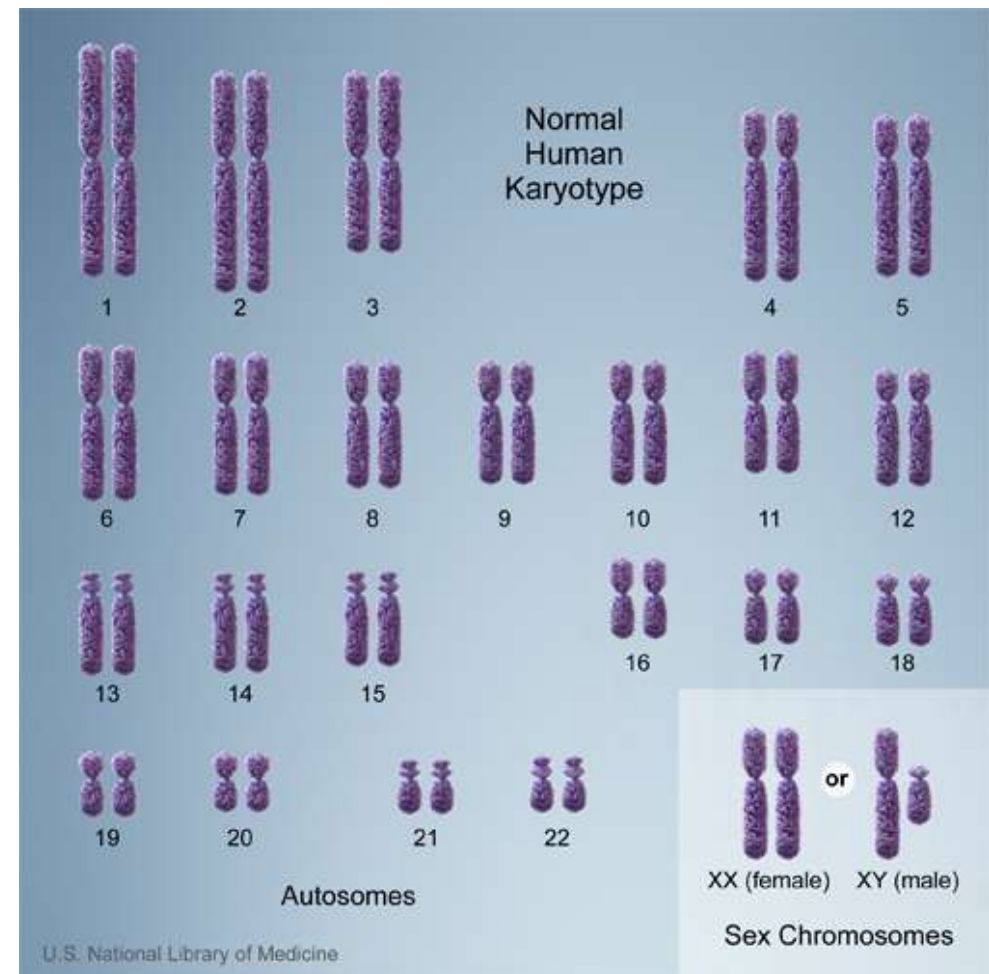
What are characteristics of the average cell?

- The average human cell has **23 pairs of chromosomes**
 - 1 from each parent, for a total of 46 per cell
- *Remember:* we have about 25,000 pairs of genes spread across our 23 pairs of chromosomes



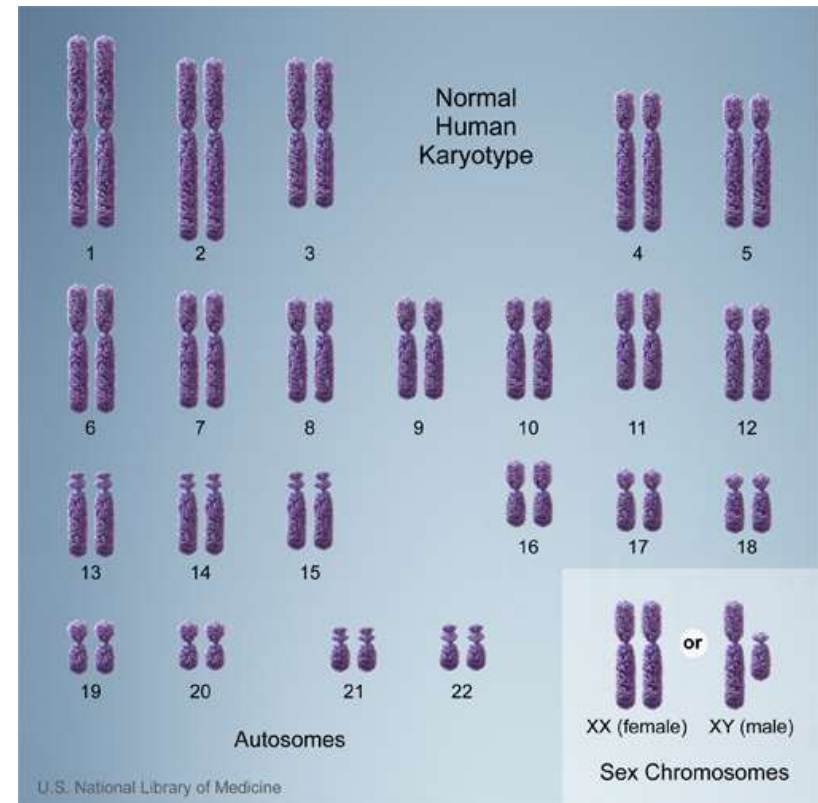
We can visualize our 23 pairs of chromosomes in a **karyotype**: a size-ordered chart of all the chromosomes in one cell

- Cells with 2 sets of chromosomes (1 set from each parent) = **diploid**
- Cells with 1 set of chromosomes = **haploid**
- 22 of our 23 chromosomes are **autosomes** (chromosomes that are the same for all people)

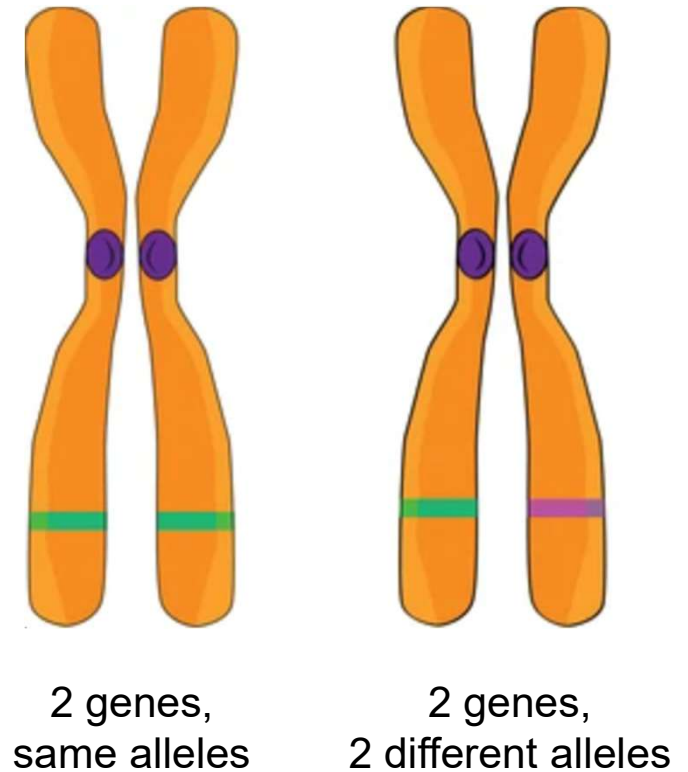


- For autosomes, each chromosome from 1 parent has a matching chromosome from the other parent = **homologous chromosomes**
 - homologous = “having the same basic structure”

- e.g. chromosome 21 includes 367 genes in a very specific order
 - this is exactly the same for both copies, 1 from each parent

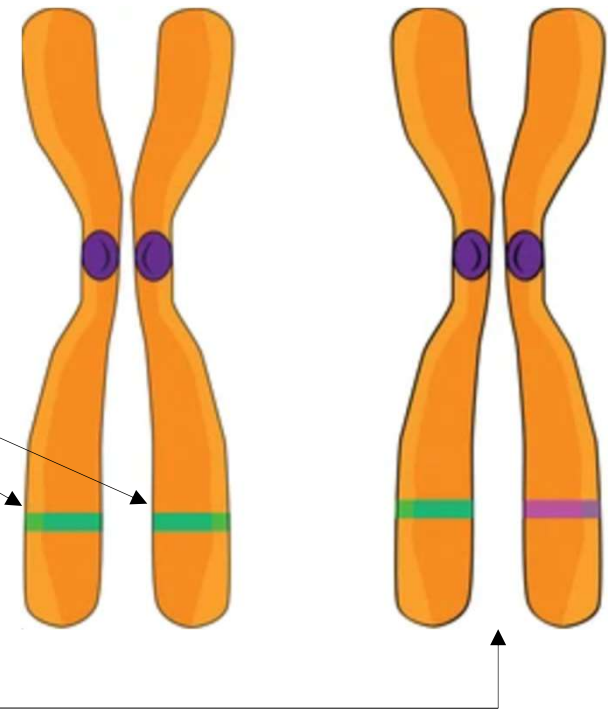


- Homologous chromosomes match even though they are from different parents
 - HOWEVER: homologous chromosomes are not necessarily identical
 - Have the same genes in the same order
 - Do NOT necessarily have the same alleles
 - **Alleles** = different versions of a gene

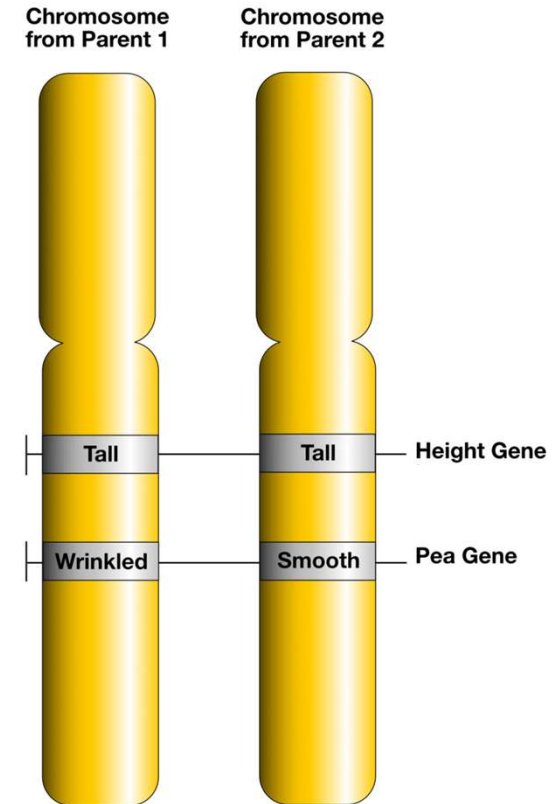


Alleles = different versions of a gene

- e.g. all of us have 2 genes for blood type, but we have some combination of the different possible alleles that code for blood type: A, B, or O
 - a person's homologous chromosomes might be identical: 1 "O" allele from each parent
 - but they might also be different: an A allele 1 parent & B allele the other



- Other allele examples:
 - cats have 2 genes for fur length, which has the possible alleles coding for short & long fur
 - e.g. plants have 2 genes for flower color, but there can be many different possible alleles, coding for red, yellow, or purple flowers



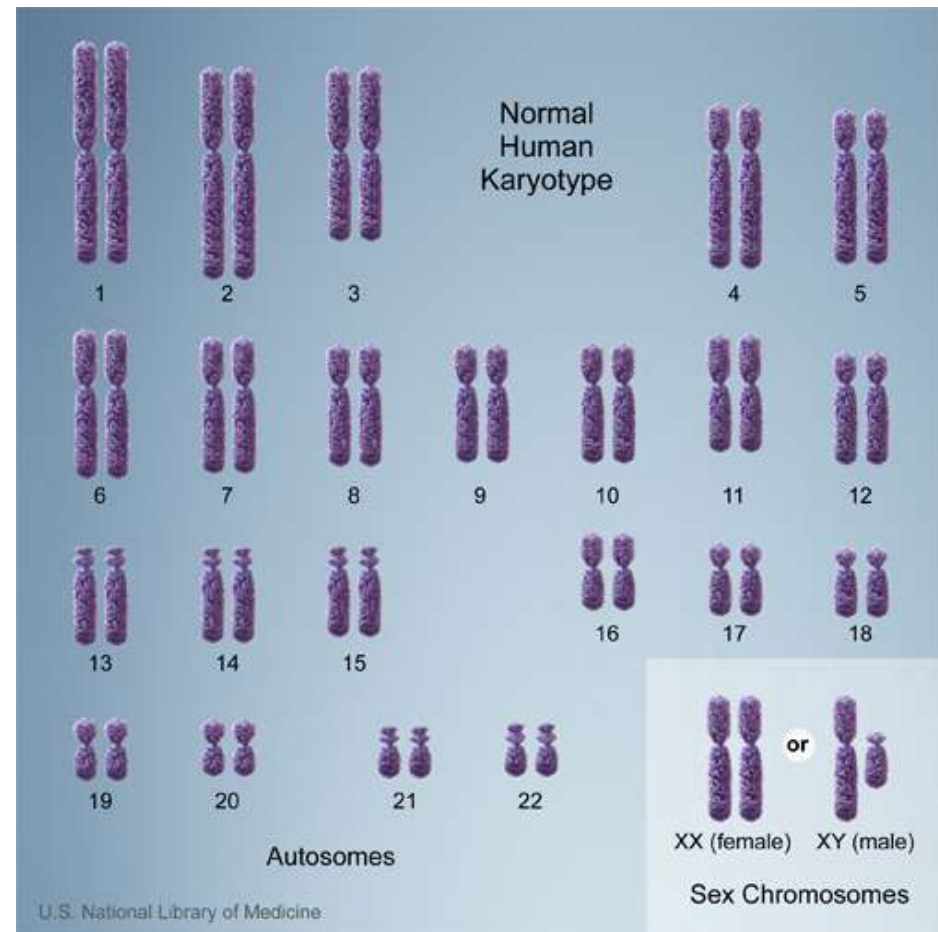
- e.g. pea plants have 2 genes for pea shape, which has the possible alleles coding for wrinkled or smooth peas

- The last pair of chromosomes = **sex chromosomes**

- these usually determine whether a person will develop:

- ovaries – X & X
or

- testes – X & Y



- Not all cells are diploid, with 2 sets of chromosomes

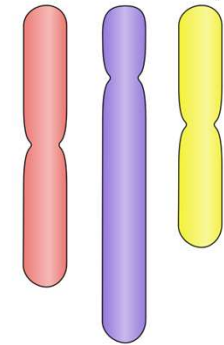
- Some cells are **haploid**: they only have 1 set of chromosomes (half)

- these cells are **gametes**:
sperm or egg cells

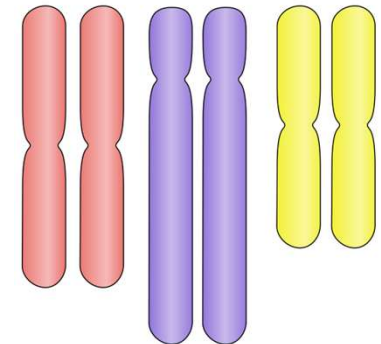
- gametes are made in ovaries (eggs) or testes (sperm) through a type cell division called **meiosis**

In egg or sperm cells (gametes), we have only one of each chromosome

Haploid (N)



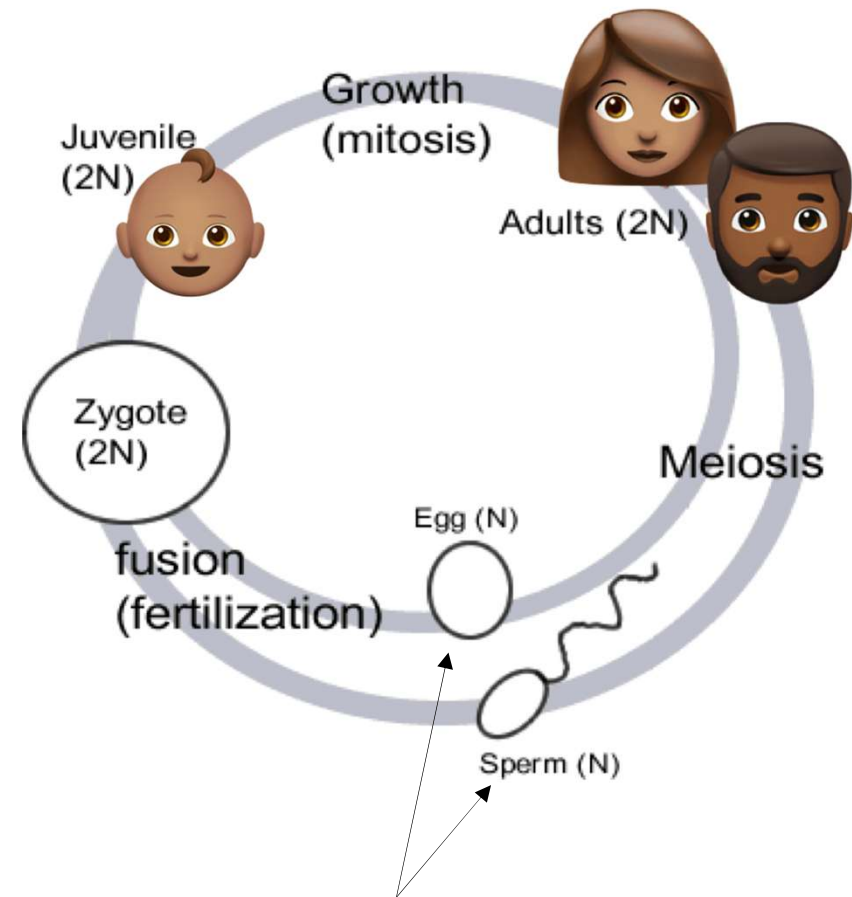
Diploid (2N)



In regular cells, we have one chromosome from each parent (two of each = diploid)

Why would we make haploid cells?

- Sexual reproduction requires the union of two specialized cells, called gametes
- It is the nuclear division that forms haploid cells from diploid cells
- 2 cells (sperm & egg) must meet to make 1 new organism
 - when a haploid sperm fertilizes a haploid egg, the offspring will have 2 sets of chromosomes: now diploid

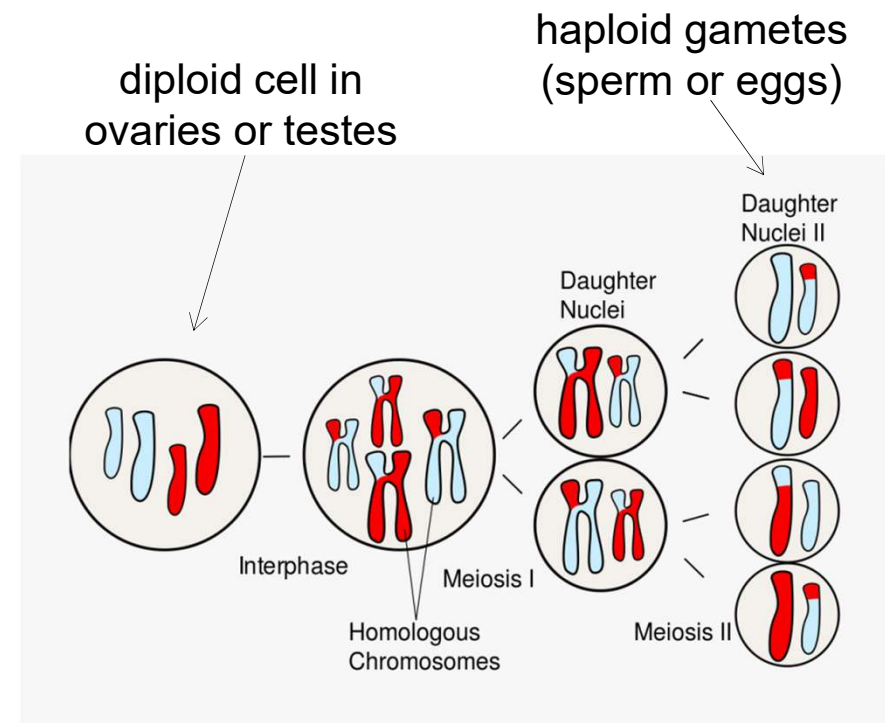


Haploid gametes fuse:
 $1N + 1N = 2N$

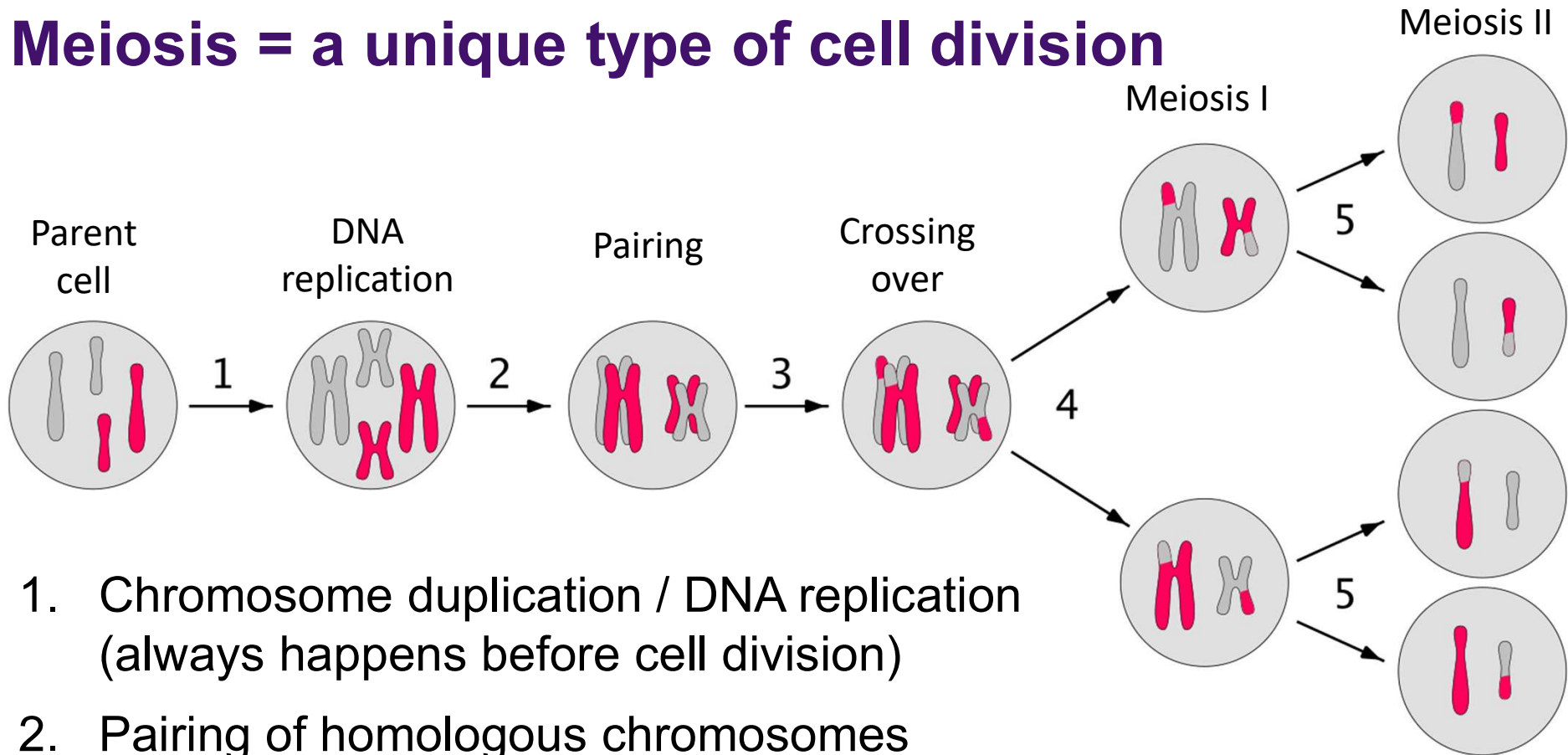
So offspring is now 2N
(diploid) like their parents

Meiosis = a unique type of cell division

- Haploid gametes for sexual reproduction are made through **meiosis**
 - Through 2 divisions, 1 parent cell produces 4 daughter cells / gametes
 - Each gamete has half the genetic material of the parent cell (haploid)
 - Gametes are NOT genetically identical to the parent cell



Meiosis = a unique type of cell division



1. Chromosome duplication / DNA replication (always happens before cell division)
2. Pairing of homologous chromosomes
3. Crossing-over (*more on this soon*)
4. Meiosis I (first division) – one of each duplicated chromosome per daughter cell
5. Meiosis II (second division) – one of each chromosome per daughter cell (gamete)

Meiosis stages

Key stages: 1. Prophase 2. Meiosis I 3. Meiosis II



Cell vision Process: 1. Prophase 2. Metaphase 3. Anaphase 4. Telophase 5. Cytokinesis

Meiosis

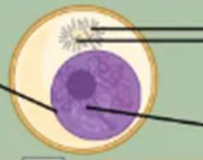
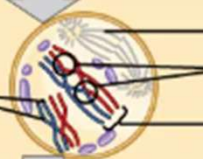
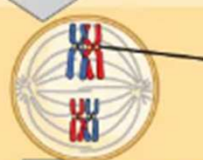
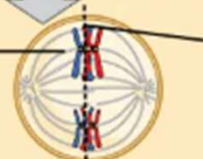
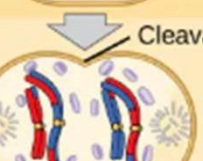
Key stages:

1. Prophase
2. Meiosis I
3. Meiosis II

Cell vision

Process:

1. Prophase
2. Metaphase
3. Anaphase
4. Telophase
5. Cytokinesis

Stage		Event	Outcome
INTERPHASE	S phase	 Centrosomes (with centriole pairs) Nuclear envelope Chromatin	Chromosomes are duplicated during interphase. The resulting sister chromatids are held together at the centromere. The centrosomes are also duplicated.
	Prophase I	 Sister chromatids Chiasmata Tetrad	Chromosomes condense, and the nuclear envelope fragments. Homologous chromosomes bind firmly together along their length, forming a tetrad. Chiasmata form between non-sister chromatids. Crossing over occurs at the chiasmata. Spindle fibers emerge from the centrosomes.
MEIOSIS I	Prometaphase I	 Centromere (with kinetochore)	Homologous chromosomes are attached to spindle microtubules at the fused kinetochore shared by the sister chromatids. Chromosomes continue to condense, and the nuclear envelope completely disappears.
	Metaphase I	 Microtubule attached to kinetochore Metaphase plate	Homologous chromosomes randomly assemble at the metaphase plate, where they have been maneuvered into place by the microtubules.
	Anaphase I	 Sister chromatids remain attached. Homologous chromosomes separate.	Spindle microtubules pull the homologous chromosomes apart. The sister chromatids are still attached at the centromere.
	Telophase I and Cytokinesis	 Cleavage furrow	Sister chromatids arrive at the poles of the cell and begin to decondense. A nuclear envelope forms around each nucleus and the cytoplasm is divided by a cleavage furrow. The result is two haploid cells. Each cell contains one duplicated copy of each homologous chromosome pair.

Meiosis

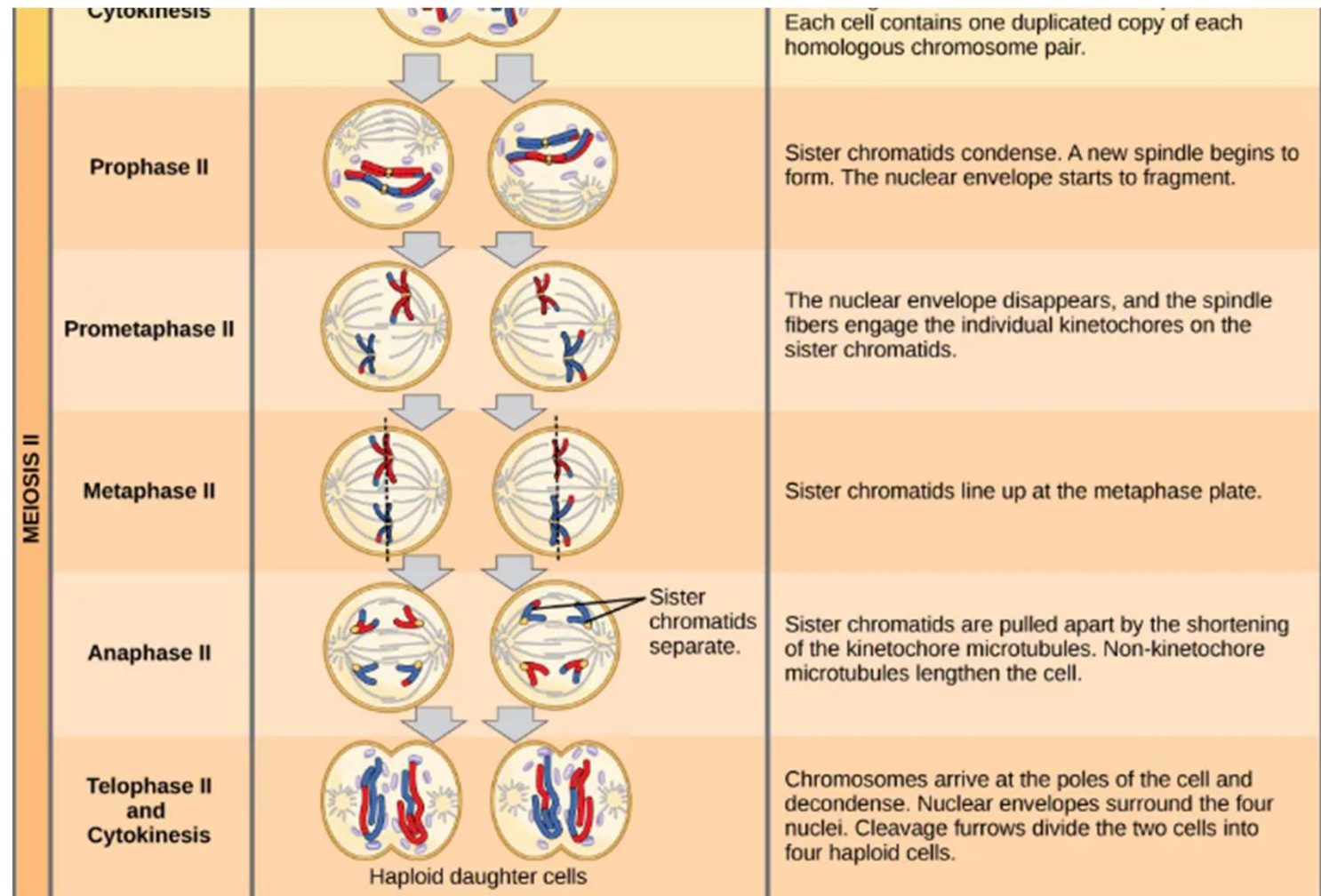
Key stages:

1. Prophase
2. Meiosis I
3. Meiosis II

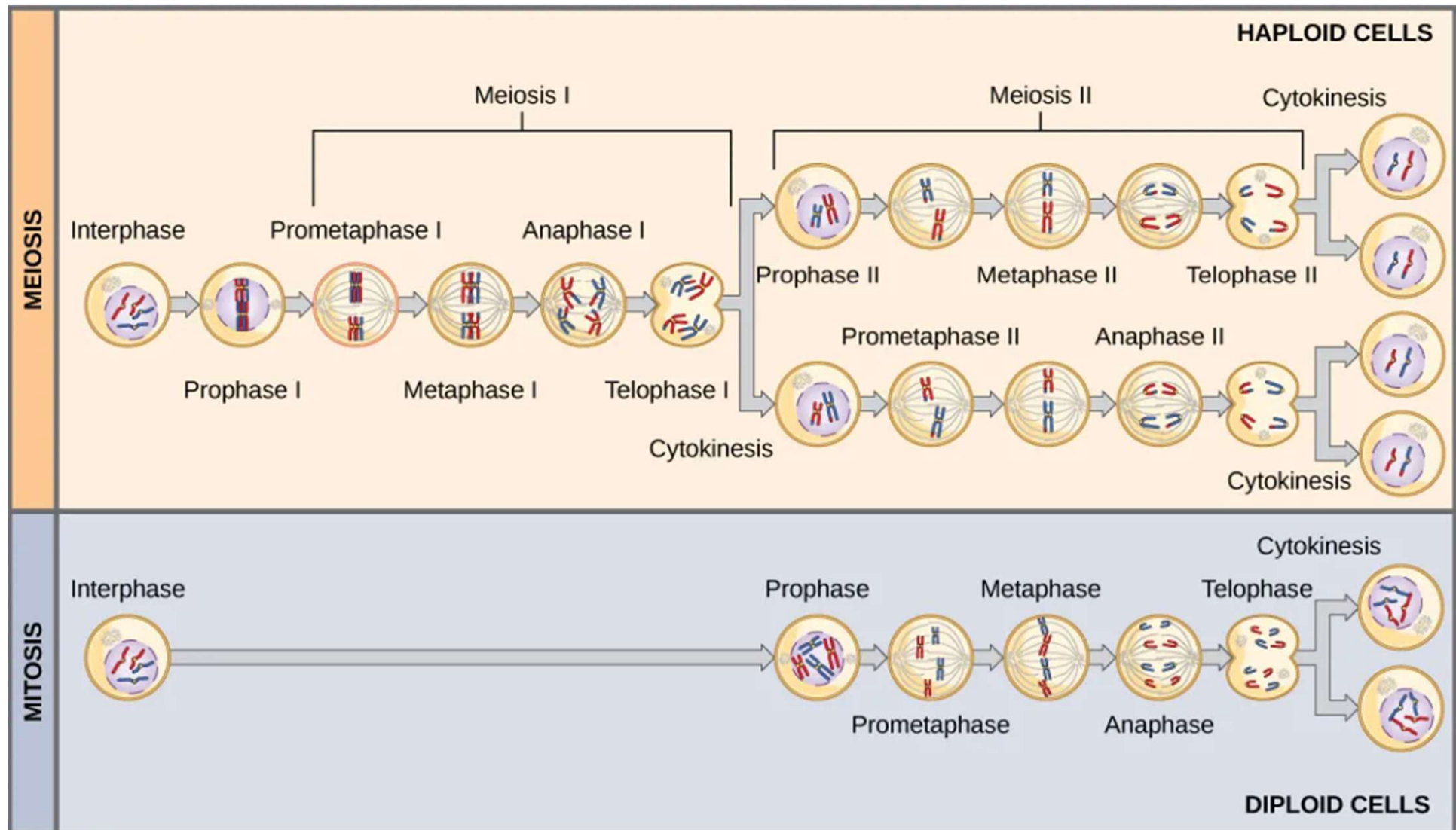
Cell vision

Process:

1. Prophase
2. Metaphase
3. Anaphase
4. Telophase
5. Cytokinesis

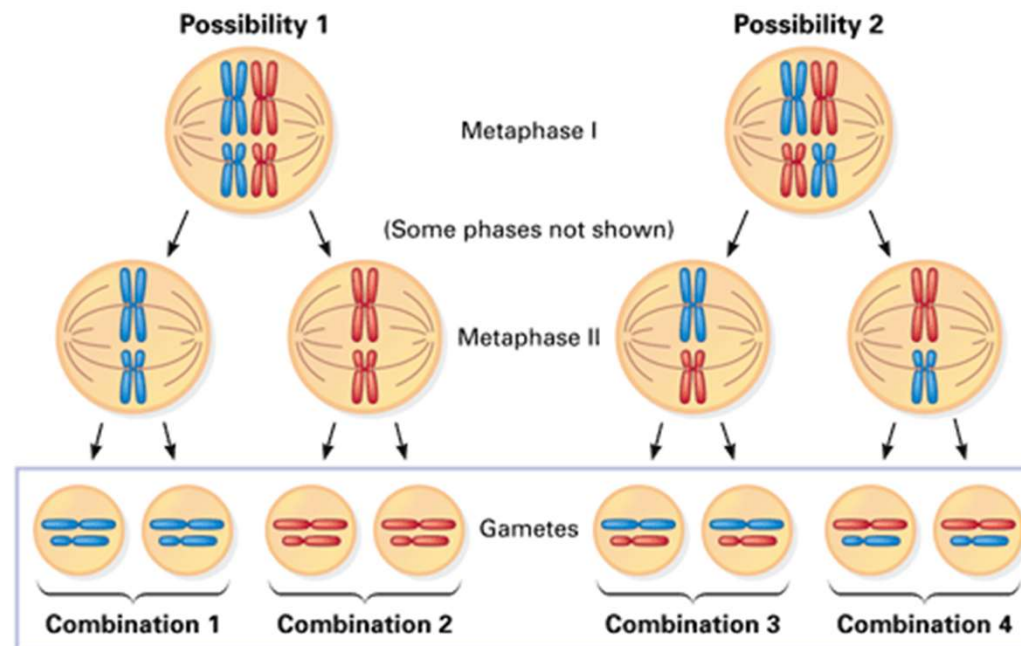


Meiosis vs Mitosis*



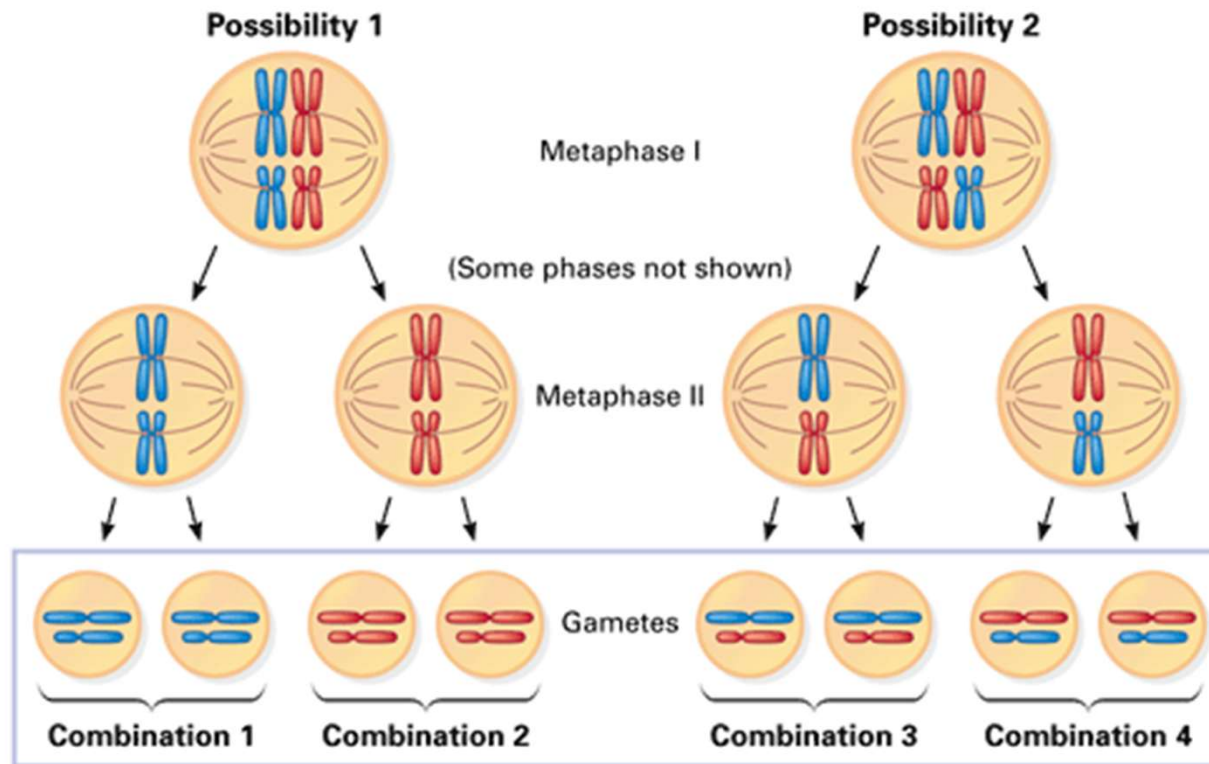
*More on mitosis in this week's lecturing

- Gametes are NOT genetically identical to each other
 - Chromosomes are randomly lined up & split during meiosis
 - this means there are several possible combinations

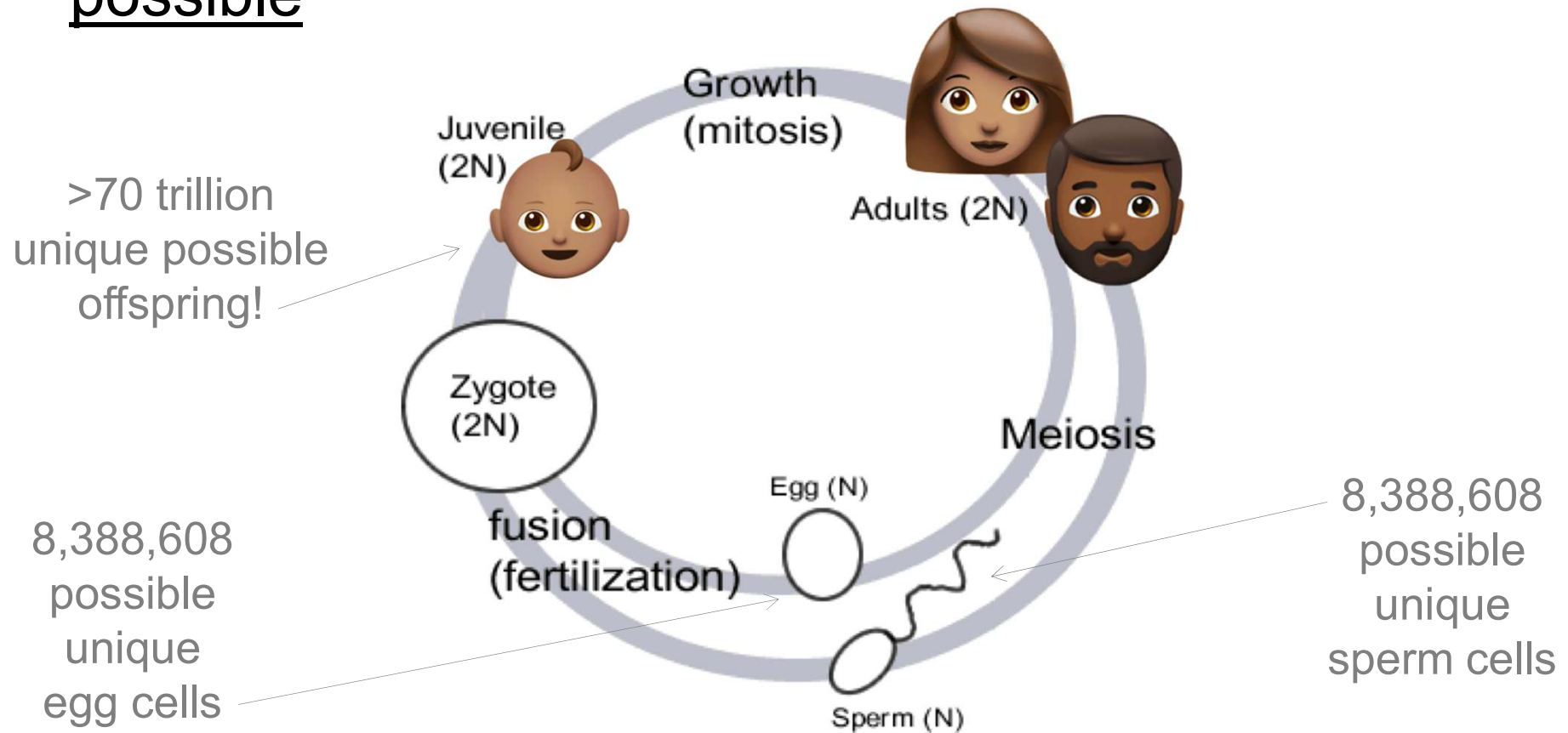


This figure shows that only one change in how two chromosomes are lined up create four possible gametes. But we have 23 chromosomes, not just two, leading to huge variation in gametes.

- Gametes are NOT genetically identical to each other
 - Humans have 23 pairs of chromosomes – when those can all line up randomly, we can make 8,388,608 possible unique gametes, all equally likely

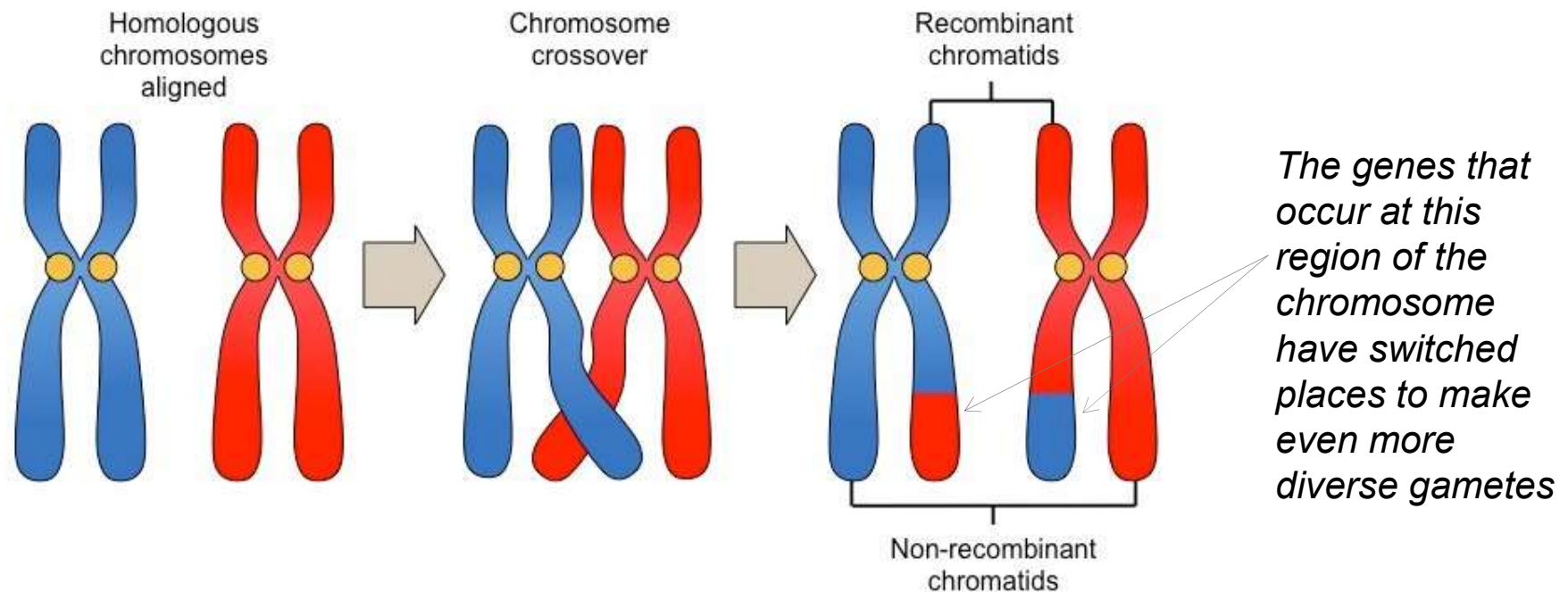


- With 23 chromosomes, incredible human diversity is possible



$8,388,608 \times 2$ (egg x sperm) = more than
70 trillion unique individuals!

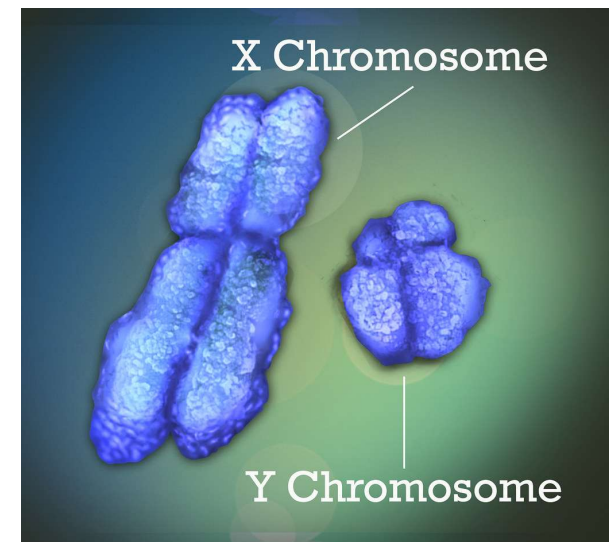
- Diversity is increased even more by crossing over
 - **Crossing over** = the exchange of DNA during meiosis – genes sometimes change places



Because of crossing over, 70 trillion unique individuals is a vast understatement!

How is sex determined in offspring?

- Mammals have a set of **sex chromosomes** that determine sex: the biological makeup of an individual's chromosomes & reproductive anatomy
 - We usually describe 2 sexes
 - Females: two X chromosomes, development of ovaries
 - Males: an X & Y chromosome, development of testes
 - *Not everyone fits into these two categories*
 - *Sex is different than gender*



- *Remember:* after meiosis each gamete only has 1 of each chromosome

- All female eggs will carry an X chromosome

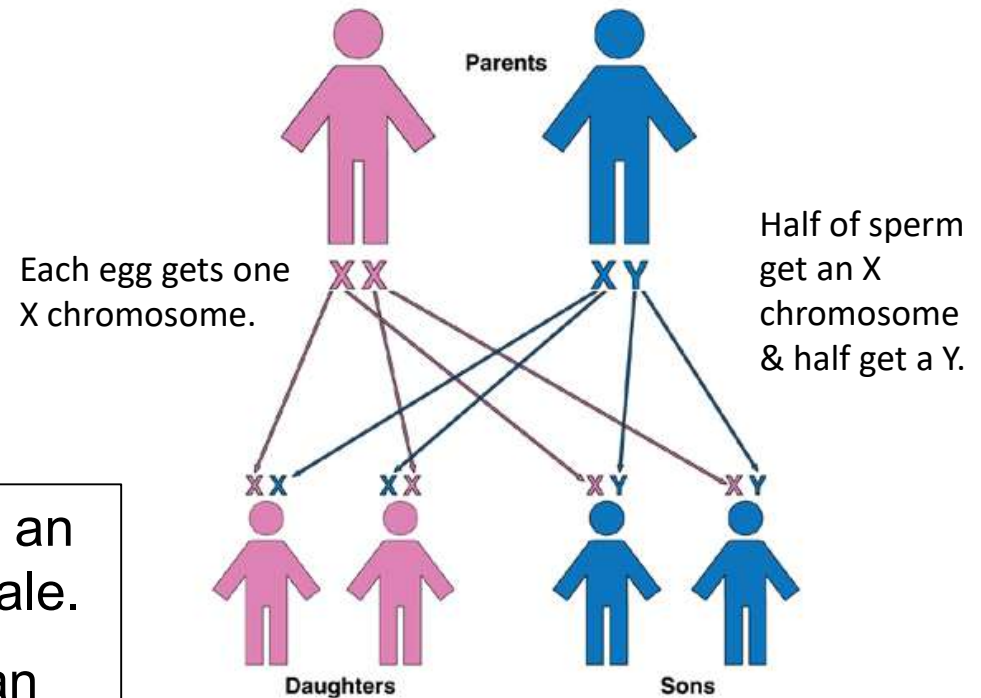
- In male sperm, half carry an X & half carry a Y

If an X-carrying sperm meets an egg, the offspring will be female.
If a Y-carrying sperm meets an egg, the offspring will be male.

Individuals have two copies of the sex chromosomes in every cell.

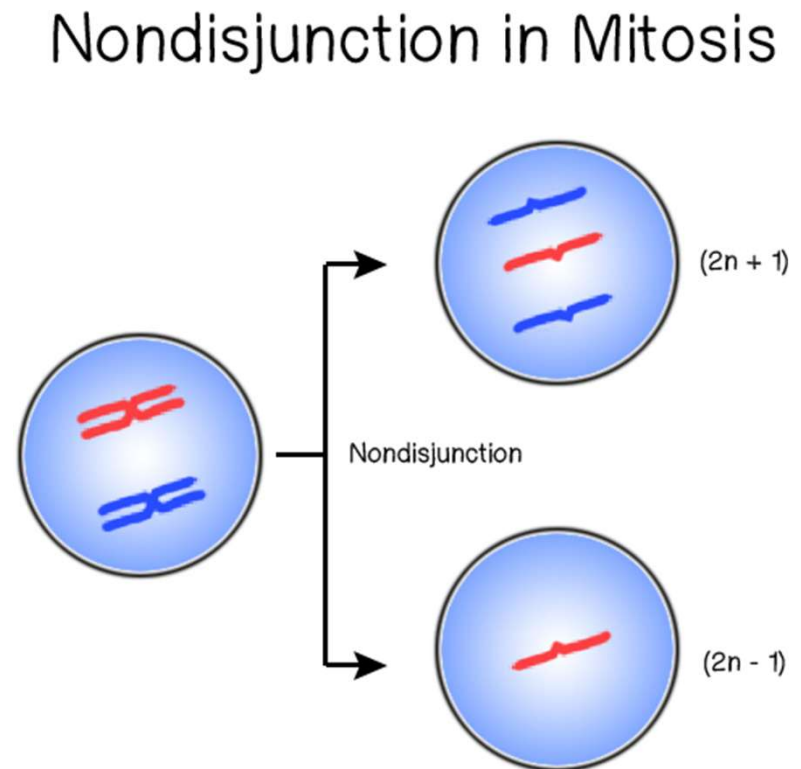
Females have two copies of the X chromosome.

Males have one X chromosome and one Y chromosome.

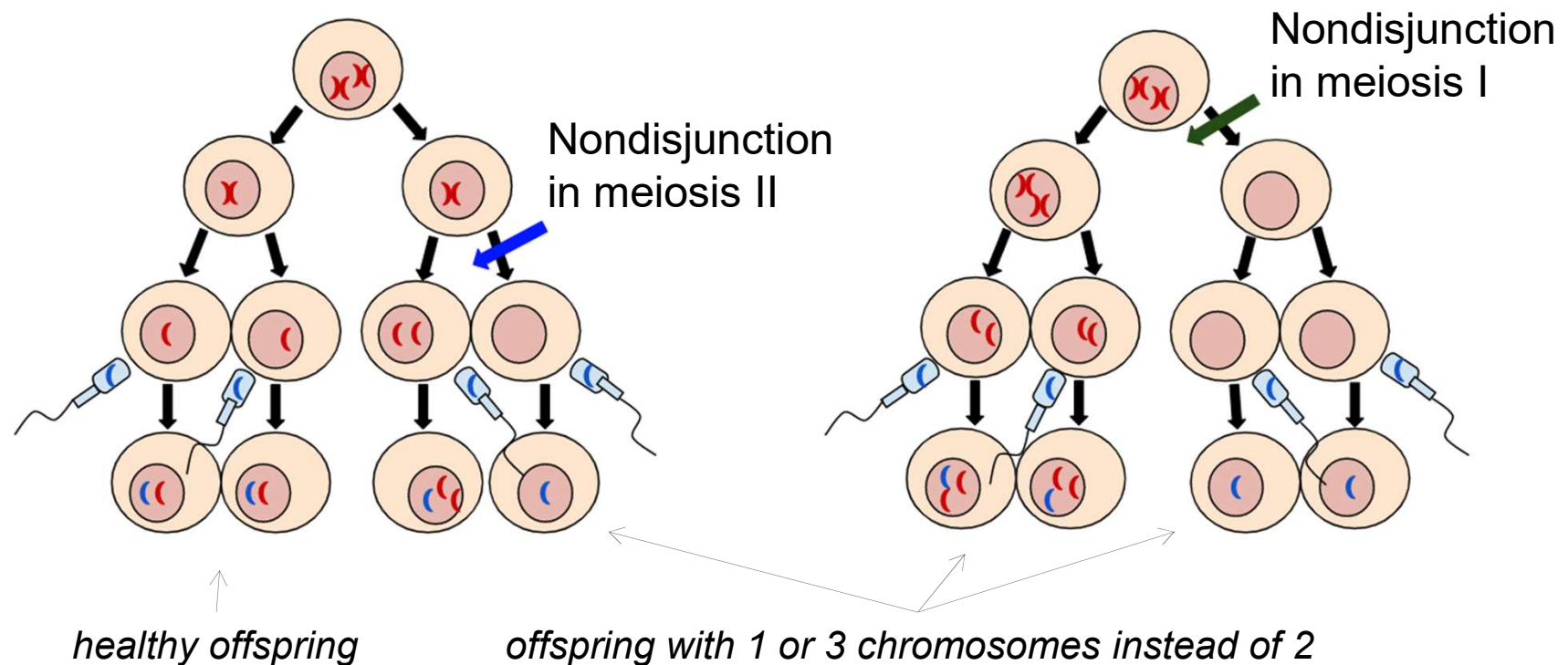


What happens when meiosis goes wrong?

- **Nondisjunction** = the incorrect separation of chromosomes at some point during meiosis
 - The result is an inappropriate number of chromosomes in the gametes (0 or 2 instead of 1)
 - Can happen in sperm or egg cells



- Nondisjunction can occur at either division in meiosis
 - *in either case, when fertilization happens the offspring will have 1 or 3 chromosomes instead of 2 (diploid)*



- What are the consequences of nondisjunction?
 - Most embryos formed from gametes with abnormal chromosome numbers result in miscarriage or die very shortly after birth
 - Some can survive: specifically if the nondisjunction happened with chromosomes 13, 18, 21, or the sex chromosomes
 - these individuals may have impairments



This person has nondisjunction in the sex chromosomes

- A common example of nondisjunction = **down syndrome**
 - Also called trisomy 21 because affected individuals have 3 copies of chromosome 21
 - includes several characteristics, including distinctive facial features & often varying degrees of learning difficulties

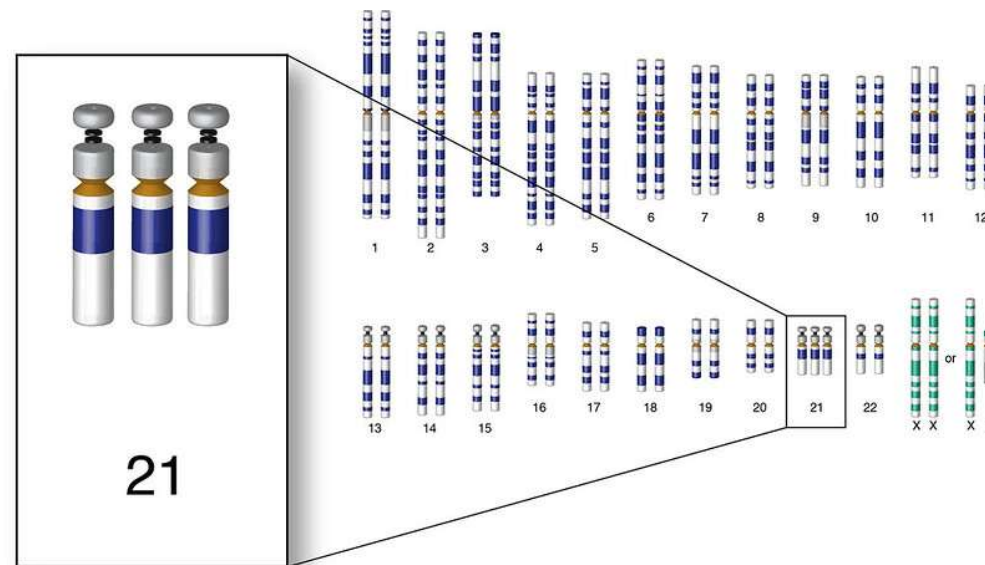


Figure: <https://www.publicdomainpictures.net/en/view-image.php?image=156019&picture=child-with-down-syndrome>

Figure: <https://www.flickr.com/photos/nihgov/28006685670>

- Other common examples of nondisjunction occur with sex chromosomes

– these can cause problems with reproduction & development but can also have no symptoms at all

Genotype	Syndrome characteristics
XO	Turner Syndrome (female, one X) • short height • underdeveloped ovaries, often infertile • hormone deficiencies can lead to prevention of puberty & menstruation
XXY	Klinefelter Syndrome (male) • underdeveloped testes • lower testosterone levels, usually infertile • possible development of some female features • however, many males show no symptoms
XYY	XYY (male) • most have normal development & offspring • more prone to tallness & acne but most males show no symptoms
XXX	XXX / Trisomy X (female) • most have normal development & offspring
YO	One Y - not viable (offspring would not survive)

- The frequency of nondisjunction in both males & females increases with age
 - Doctors will often run more tests on pregnant females older than 35 to find out early about nondisjunction

